

INTREPRETATION

G Kushwanth 192424150

1. An e-commerce company is analysing the purchase amounts of 10,000 customers during a festive sale. Most customers spent between ₹500 and ₹2,000, while a small number of customers made purchases above ₹25,000. The company wants to understand customer spending behaviour, identify high value customers, and determine whether the spending distribution follows a normal pattern. How would you use ECDF, histograms, and Q-Q plots to analyse the spending data, and what insights could be derived from the results?

The company wants to understand customer spending behaviour, identify high value customers, and determine whether the purchase amounts follow a normal distribution.

1. Histogram

A histogram groups purchase amounts into intervals and shows the frequency of customers in each interval.

- Most bars will be concentrated between **₹500 and ₹2,000**, indicating that most customers made moderate purchases.
- A few bars at the higher end (above **₹25,000**) indicate the presence of high value customers (outliers).
- The histogram also reveals whether the spending distribution is symmetric or positively skewed.

2. ECDF (Empirical Cumulative Distribution Function)

The ECDF displays the cumulative percentage of customers whose spending is less than or equal to a given value.

- It helps determine the percentage of customers spending below a specific amount.
- For example, the ECDF may show that nearly **95% of customers spent below ₹2,000**, while only a very small percentage spent above ₹25,000.
- This visualization is useful for identifying customer spending percentiles.

3. Q-Q Plot

A Q-Q Plot compares the customer spending data with a theoretical normal distribution.

- If the plotted points lie close to the reference line, the data is approximately normally distributed.
- Large deviations from the line, especially at the upper end, indicate that the spending distribution is **not normal** due to a few extremely high purchase amounts.

Insights

- Most customers make purchases between **₹500 and ₹2,000**.
- A small number of customers contribute significantly higher purchase values.
- Customer spending is **positively skewed** because of high-value purchases.
- The company can identify premium customers for loyalty programs and personalized marketing.

Conclusion

Histogram, ECDF, and Q-Q Plot together provide a comprehensive understanding of customer spending behaviour. They help identify outliers, evaluate data distribution, and support business decisions for customer segmentation and targeted promotions.

2.A hospital is comparing patient waiting times in two departments: Emergency Care and General Consultation. The management wants to determine whether waiting times in both departments follow similar distributions and whether one department experiences significantly longer delays. Explain how Q-Q plots, boxplots, and density plots can be used to compare the two datasets and evaluate distributional differences.

A hospital compares patient waiting times in **Emergency Care** and **General Consultation**. Explain how **Q-Q Plot**, **Boxplot**, and **Density Plot** help compare the distributions and identify longer waiting times.

Answer

The hospital aims to compare patient waiting times across two departments and

determine whether one department experiences significantly longer delays.

1. Boxplot

The Boxplot summarizes the waiting-time distribution using the median, quartiles, and outliers.

- The department with a higher median has longer waiting times. •

A wider box indicates greater variability.

- Individual points outside the whiskers represent unusually long waiting times.

2. Density Plot

The Density Plot provides a smooth representation of waiting-time distributions.

- It allows direct comparison between Emergency Care and General Consultation.
- Peaks represent the most common waiting times.
- Wider curves indicate greater variation in waiting times.

3. Q-Q Plot

The Q-Q Plot compares waiting-time data with a theoretical normal distribution.

- If points follow the reference line, waiting times are approximately normally distributed.
- Deviations from the line suggest skewness or the presence of outliers.

Analysis

- Emergency Care generally experiences longer waiting times because of critical patient cases.
- General Consultation usually has shorter and more consistent waiting times. •

Outliers may represent emergency situations requiring extended treatment.

Conclusion

Boxplots, Density Plots, and Q-Q Plots effectively compare waiting-time distributions, identify variability, detect outliers, and assist hospital management in improving patient scheduling and reducing delays.

3.A digital marketing team wants to present the number of likes received by five social media campaigns: 2,000, 5,000, 8,000, 15,000, and 30,000 likes. One analyst suggests using a 3D exploded pie chart, while another recommends a horizontal bar chart. Discuss how each visualization may influence audience perception and identify the most effective method for accurately comparing campaign performance. What design principles should be followed to avoid misleading interpretations?

A marketing team compares a **3D Exploded Pie Chart** and a **Horizontal Bar Chart** to display likes received by five social media campaigns. Explain which visualization is more effective and justify your answer.

Answer

The marketing team wants to compare likes received by five

campaigns: • Campaign A – 2,000

• Campaign B – 5,000

• Campaign C – 8,000

• Campaign D – 15,000

• Campaign E – 30,000

1. 3D Exploded Pie Chart

Advantages:

- Attractive and visually appealing.
- Highlights one particular category.

Disadvantages:

- Difficult to compare similar values accurately.
- Three-dimensional effects distort slice sizes.
- Can mislead viewers about actual proportions.

2. Horizontal Bar Chart

Advantages:

- Provides accurate comparison of campaign performance.
- Easily identifies the highest and lowest values.
- Suitable when category labels are long.
- Easy to interpret and compare differences.

Best Visualization

The **Horizontal Bar Chart** is the most effective visualization because it accurately compares campaign performance without introducing visual distortion. **Design**

Principles

To avoid misleading interpretations:

- Use consistent scales.
- Avoid unnecessary 3D effects.
- Clearly label axes and categories.

- Use simple and readable colors.
- Arrange bars in ascending or descending order.
- Include appropriate titles and legends.

Conclusion

Although a 3D exploded pie chart is visually attractive, a horizontal bar chart provides a more accurate and reliable comparison of campaign performance. Therefore, the horizontal bar chart is the preferred visualization for decision making and presentation.